

# Youse Garcia

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## EDUCATION

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### University of Connecticut

Storrs, CT

*Bachelor of Science in Mechanical Engineering, Aerospace concentration*

*Jan. 2025 – May 2029*

- **Cumulative GPA: 2.9** — **Major Core GPA: 3.3**

## EXPERIENCE

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### NASA L'SPACE NPWEE Program

Remote

*Participant / Technical Team Member*

*Sept. 2026 – Present*

- Selected and participating in the NASA Proposal Writing and Evaluation Experience (NPWEE), collaborating in a multidisciplinary team with undergraduates and industry professionals to develop a project following NASA protocols.

### NASA L'SPACE Mission Concept Academy

Remote

*Mechanical Engineer – Rover Design Team*

*Aug. 2025 – Jan. 2026*

- Designed chassis, solar layout, and 4-wheel drive mobility for a Martian rover while managing component budgets, size envelopes, and material selection based on thermal/structural targets.
- Conducted trade studies evaluating in-house manufacturing versus vendor procurement and lead times.
- Integrated mechanical hardware with thermal, electrical, and science payloads while completing formal NASA reviews (**MCR, SRR, PDR, MDR**).

### Clarkson University

Potsdam, NY

*Undergraduate Research Assistant*

*June 2024 – Sept. 2024*

- Researched powered prosthetic limbs and electromechanical systems under Dr. Kevin Fite, supporting prosthetic hand prototype design and mechanical function replication.
- Authored and presented project findings at Clarkson's Research Showcase (**RAPS**) and the national Biomedical Engineering Society (**BMES**) Conference.

### Clarkson University

Potsdam, NY

*Undergraduate Research Assistant*

*Sept. 2024 – Dec. 2024*

- Performed pipetting, PCR, gel electrophoresis, breeding, and microscope analysis while managing daily lab upkeep in Dr. Wallace's zebrafish laboratory

### Clarkson University

Potsdam, NY

*Facilities and Services Intern*

*May 2025 – Aug. 2025*

- Surveyed **20+** campus buildings and mapped **1,000+** fire safety devices into Clarkson's GIS system, boosting operational efficiency.
- Reviewed blueprints for **500+** fire doors and inspected **200+** windows to verify fire code compliance and streamline equipment replacement.
- Utilized GIS Pro and PTGui to scan **400+** rooms and build 3D facility models for virtual campus tours.

## PROJECTS

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### Martian Exploration Bridge Design — *SOLIDWORKS, 3D CAD, Prototyping*

Spring 2026

- Designed a Martian exploration bridge from initial sketches to a final 3D CAD model within strict size and budget constraints.
- Integrated rover clearance, internal LED routing, and a pedestrian walkway; presented final documentation at the UConn First-Year Engineering Expo.

## TECHNICAL SKILLS & AFFILIATIONS

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**CAD & Design:** SOLIDWORKS, Siemens NX (Certification In Progress), PTGui, ArcGIS Pro, Blueprint Reading  
**Engineering Methods:** Formal Design Reviews (**MCR, SRR, PDR, MDR**), Trade Studies, Material Selection, Budget Management

**Affiliations:** Borealis Scholar Alum (NIH-Funded Research Initiative), AIAA (Clarkson Chapter), SHPE (Clarkson Chapter)